

SPATIAL ESTIMATION OF SURFACE URBAN HEAT ISLAND INTENSITY IN DAVAO CITY USING GRAPH NEURAL NETWORKS AND ROAD TOPOLOGY



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CHAPTER I

INTRODUCTION

Introduction

Rapid urbanization and changes in land-use patterns have intensified global environmental challenges, particularly in megacities facing the dual pressures of densification and climate change [1]. As the human population grows, urban residents face increasingly extreme temperatures that threaten well-being and economic stability [2]. This is especially critical in the Global South, where heatwaves disrupt labor productivity, strain energy systems, and disproportionately affect vulnerable communities [3].

In the Philippines, the combination of high humidity and an archipelagic climate makes the country uniquely susceptible to these thermal risks, with historical analyses showing significant warming trends driven by unchecked urbanization [4], [5]. Recent nationwide observational studies confirm that accelerating development has substantially altered local thermal profiles, particularly elevating minimum temperatures across major urban centers [6]. Furthermore, atmospheric modeling reveals that dense structural surfaces and anthropogenic heat emissions account for severe nocturnal heat retention in Philippine cities [7].

This regional trend is acutely observable in Davao City, the primary socio-economic hub of Southern Mindanao. Local environmental monitors have already classified the city's urban heat index as above normal, a condition attributed to the rapid conversion of green spaces into concrete infrastructure [8]. Previous studies in Davao have identified that residents in the main urban districts already experience the warmest temperatures, driven largely by building density [9]. Recent spatiotemporal analyses further indicate that commercial expansion across Davao's congressional districts has drastically diminished green space, driving Land Surface Temperatures (LST) upward—most notably within densely built-up, vegetation-poor barangays [10]. Given the scarcity of dense meteorological sensor networks in Davao, relying on Surface Urban Heat Island (SUHI) intensity via satellite remote sensing remains the most effective metric for spatially explicit urban planning [11].

Despite the growing urgency to diagnose local thermal stress, existing approaches to SUHI modeling suffer from significant methodological gaps. Traditional environmental remote sensing relies heavily on pixel-based structures analyzed via conventional Convolutional Neural Networks (CNNs) or global statistical classifiers like Random Forest (RF) regression [12], [13]. However, regular Euclidean grid structures fail to capture the complex, non-Euclidean geometry of urban morphology [14]. Furthermore, standard machine learning models like RF assume spatially uniform relationships, thereby blurring fine-grained spatial heterogeneity across localized microclimates [15]. Modeling urban networks using naive Euclidean distances creates fundamental distortion because microclimatic heat propagation operates along topological structures—such as road networks, building alignments, and street canyons [16].

To overcome these structural limitations, recent advancements in deep learning favor graph-based architectures [17]. Spatial Graph Convolutional Networks (GCNs) map urban environments as non-Euclidean graphs—treating road intersections as nodes and connecting segments as edges—thereby capturing spatial dependencies that grid-based models miss [18], [19]. Furthermore, by integrating node positions and structural topology, Spatial GCNs can effectively construct feature maskings that generalize better than standard image processing techniques [20]. GCNs have emerged as a superior framework for environmental modeling, specifically because they capture context-specific urban heat patterns (such as the buffering effects of vegetation versus heat retention of asphalt) that ordinary feature-based models overlook [21]. Additionally, graph representation learning has proven exceptional at estimating continuous spatial LST fields and handling cloud-contaminated datasets [22] [23].

While intra-urban heat diagnostics have demonstrated success in identifying high-risk thermal zones in Manila [24], Davao City currently lacks a topologically grounded, high-resolution microclimatic estimation model. This study addresses this critical gap by integrating OpenStreetMap (OSM) road network topology with multi-source satellite imagery through a Spatial Graph Convolutional Network to accurately model and estimate SUHI intensity across Davao City.

1.2 Statement of the Problem

The primary purpose of this research is to develop a Spatial Graph Convolutional Network (GCN) model to spatially estimate and map Surface Urban Heat Island (SUHI) intensity in Davao City. Unlike traditional remote sensing approaches that treat the urban environment as a collection of isolated pixels, this study aims to model the city as a connected graph, leveraging road network topology to understand how heat radiates and flows between adjacent intersections.

Specifically, this study aims to answer the following research questions:

- How can the integration of OpenStreetMap (OSM) road network topology with satellite thermal imagery (Landsat 8/9 and ECOSTRESS) improve Surface Urban Heat Island (SUHI) estimation accuracy in Davao City?
- Can the proposed Spatial GCN framework accurately identify and localize high-risk thermal zones at the street level, particularly in areas with a high density of impervious surfaces?
- How does the predictive accuracy of the proposed Spatial GCN model compare to a baseline Random Forest (RF) regression model, as evaluated by standard regression metrics (RMSE, MAE, and R^2)?
- How do thermal mitigating factors, such as vegetation density (NDVI), influence and buffer the predicted heat distribution within the spatial graph network?

1.3 Objectives of the Study

The main purpose of this research is to develop a Spatial Graph Convolutional Network (GCN) model to spatially estimate and map Surface Urban Heat Island (SUHI) intensity in Davao City.

To achieve this primary goal, the study aims to fulfill the following specific objectives:

- To evaluate how the integration of OpenStreetMap (OSM) road network topology with satellite thermal imagery (Landsat 8/9 and ECOSTRESS) improves Surface Urban Heat Island (SUHI) estimation accuracy in Davao City.

- To assess whether the proposed Spatial GCN framework can accurately identify and localize high-risk thermal zones at the street level, particularly in areas with a high density of impervious surfaces.
- To compare the predictive accuracy of the proposed Spatial GCN model against a baseline Random Forest (RF) regression model using standard regression metrics (RMSE, MAE, and R2).
- To investigate how thermal mitigating factors, such as vegetation density (NDVI), influence and buffer the predicted heat distribution within the spatial graph network.

1.4 Significance of the Study

This study develops a spatial estimation framework for urban thermal monitoring, demonstrating the feasibility of using Earth observation data and graph-based machine learning to provide granular heat risk intelligence. The findings will benefit the following sectors:

- **Urban Planning and Engineering:** Provides the City Planning and Development Office (CPDO) and the City Engineer's Office (CEO) with empirical data to assist in infrastructure development. Identifying specific road corridors as "heat traps" serves as a reference for evaluating cool pavement technologies and permeable concrete in high-heat zones.
- **Local Government and Disaster Management:** Offers a data-driven, scientific reference for the City Council to review Zoning Ordinances and the Comprehensive Land Use Plan (CLUP). Instead of a one-size-fits-all approach, it pinpoints the most effective locations for new tree planting and green corridors.
- **Technological and Scientific Contribution:** Advances the field of Computer Science by establishing a reproducible methodological framework for integrating non-Euclidean data (road networks) with raster-based satellite imagery using open-source data as a cost-effective alternative to proprietary models.
- **Societal and Community Impact:** Highlights the critical issue of thermal comfort and walkability for the general public, fostering community support for pedestrian infrastructure improvements like covered walkways and street-level shading.

1.5 Scope and Delimitations

To maintain methodological rigor, this research is bounded by the following parameters:

- **Geographical Scope:** While Davao City encompasses a vast administrative land area, this study spatially delimits its analysis strictly to the Davao City Urban Core (Downtown Commercial and Dense Residential Districts)—specifically encompassing Poblacion, Agdao, and the contiguous built-up areas of Buhangin and Talomo. This boundary explicitly excludes outlying agricultural, mountainous, and rural districts (such as Marilog, Paquibato, and Baguio) to ensure that the spatial graph network $G = (V, E)$ maintains high topological connectivity and focuses exclusively on urbanized street canyons where impervious surface density (NDBI) actively drives Surface Urban Heat Island (SUHI) intensity.
- **Temporal Scope:** Data collection focuses on the drier seasons (February to April) from 2023 to 2025 to capture peak thermal conditions and maximum urban heat stress. Targeting this specific seasonal window actively avoids the heavy rainfall and satellite-blocking cloud contamination that historically begins in May across the Davao region.
- **Data Sources:** The study relies exclusively on remote sensing and open-source structural data. Thermal variables will be derived from Landsat 8/9 OLI/TIRS and ECOSTRESS Level-2 products. Topological road network data will be extracted from OpenStreetMap (OSM).
- **Theoretical Delimitation:** This research focuses strictly on estimating Surface Urban Heat Island (SUHI) intensity via Land Surface Temperature (LST). It explicitly excludes the measurement or prediction of ambient air temperature (traditional atmospheric UHI) from ground meteorological stations.
- **Methodological Delimitation:** The comparative analysis is limited to evaluating the proposed Spatial GCN against a baseline Random Forest (RF) regression model to determine if topological data improves estimation accuracy over a standard, non-structural spatial algorithm.

CHAPTER II

2.1 Review of Related Literature

To provide a robust foundation for spatially estimating Surface Urban Heat Island (SUHI) intensity, this review synthesizes recent literature across four key domains: the broad impacts of urbanization on the Philippine microclimate, the specific thermal environment of

Davao City, the limitations of traditional spatial modeling, and the emergence of Graph Neural Networks (GNNs) for environmental analysis.

Urbanization and the Microclimate in the Philippines

Rapid urbanization is a primary driver of global environmental change, pushing land-use patterns toward dense, heat-retaining infrastructure that intensifies local heat islands [1], [2]. In the Global South, these thermal risks severely strain economic and health systems [3]. The Philippines, given its archipelagic climate and high humidity, is exceptionally vulnerable. Historical climate data indicates a long-term warming trend in extreme daily precipitation and near-surface air temperatures [4], [5]. Recent nationwide observational studies confirm this, demonstrating that rapid urbanization has fundamentally altered surface air temperature trends across the country, particularly elevating minimum temperatures and the diurnal temperature range [6].

In hyper-dense Philippine megacities, the replacement of natural terrain with urban canopies significantly exacerbates these conditions. Modeled attribution studies have shown that urban land cover and anthropogenic heat emissions account for a roughly 10% increase in nocturnal heat intensity during extreme heat events [7]. Consequently, intra-urban heat mapping has become a critical diagnostic tool, successfully used in Manila and Quezon City to link land surface temperatures with vegetation and built-up indicators to identify zones requiring immediate intervention [25][23].

Land Surface Temperature (LST) and SUHI in Davao City

While atmospheric Urban Heat Islands (UHI) rely on scarce meteorological sensors [13], Surface Urban Heat Islands (SUHI) utilize satellite remote sensing to provide a spatially explicit metric of how urban materials radiate heat [11], [26]. This distinction is crucial for Davao City. Local monitors report an "above normal" urban heat index driven by rampant infrastructure development [8]. Previous spatiotemporal analyses in Davao indicate that the warmest urban effects are highly localized in dense districts like Poblacion and Agdao, correlating strongly with the Normalized Difference Built-up Index (NDBI) [9].

Recent localized evidence further underscores this crisis. Davao's urban expansion has rapidly reduced natural cooling sinks, pushing LST upward across its congressional districts and pinpointing densely built-up, vegetation-poor barangays as the hottest zones [11]. Conversely, geospatial studies consistently confirm that urban green spaces and street-level vegetation act as vital thermal buffers capable of mitigating this radiated heat [27] [28].

Limitations of Traditional Spatial Modeling

While satellite imagery provides excellent spatial coverage, standard machine learning techniques for SUHI applications remain structurally limited [12]. Conventional Convolutional Neural Networks (CNNs) treat satellite data as a regular grid of Euclidean pixels, a structure that fundamentally mismatches the irregular spatial dependencies of urban road networks [14].

Furthermore, traditional baseline models like Random Forest (RF) regression are typically executed as global models. Consequently, they blur spatially varying relationships, failing to account for the spatial heterogeneity that defines microclimates [15]. When off-the-shelf models construct spatial graphs based purely on Euclidean distances, they often miss the actual topological geometry—such as road connectivity and street canyons—that drives spatial autocorrelation [16]. Therefore, treating the urban fabric merely as a rasterized grid destroys its structural integrity and fails to capture the idiosyncratic relations of road networks [19].

Graph Neural Networks in Environmental Modeling

To overcome the limitations of Euclidean grids, recent research has shifted toward non-Euclidean Graph Neural Networks [29][30]. Spatial Graph Convolutional Networks (GCNs) have proven highly successful in spatiotemporal interpolation [31], traffic forecasting [17] [32], and urban air pollution prediction [33].

In the specific domain of thermal modeling, GCNs represent a significant paradigm shift. A novel GCN-based framework demonstrated superior performance over traditional baselines by capturing spatially dependent, context-specific thermal patterns—such as the

daytime importance of vegetation versus nighttime building density—which ordinary feature-based models overlook [20]. Furthermore, graph representation learning improves the retrieval and interpolation of LST by selecting optimal thermal infrared bands and managing temporal data gaps caused by cloud cover [22][23][35]. Adaptive inductive graph representation learning has specifically outperformed standard kriging-style baselines, proving that graph-based neighborhood selection is ideal for continuous temperature estimation [22].

2.2 Conceptual Framework

The conceptual framework of this study relies on treating Davao City not as a flat image, but as an interconnected topological structure. The framework operates by fusing physical environmental data derived from remote sensing with the structural connectivity of the urban road network.

Graph Definition and Topology

The physical urban structure is mathematically formalized as a spatial graph $G = (V, E)$

- **Nodes (V):** Represent individual road intersections extracted from OpenStreetMap (OSM). Each node acts as a discrete point of localized thermal measurement.
- **Edges (E):** Represent the structural road segments connecting these intersections. Edge attributes include segment length, road classification, and estimated traffic capacity, which dictate the structural connectivity between nodes.

Node Feature Extraction

To construct an Attributed Graph, each node $v \in V$ is assigned a specific feature vector based on remote sensing indices mapped to its geographic coordinates. The target feature is the Land Surface Temperature (LST), derived from Landsat 8/9 and ECOSTRESS imagery. The input predictive features acting as the physical drivers of the microclimate include:

- **Thermal Generators:** Normalized Difference Built-up Index (NDBI), representing the density of impervious, heat-absorbing concrete and asphalt.
- **Thermal Buffers:** Normalized Difference Vegetation Index (NDVI), representing dense vegetation and tree canopy coverage.

Spatial GCN Mechanism

Once the topological graph G and the node features are initialized, the Spatial Graph Convolutional Network utilizes an adjacency matrix to aggregate thermal and physical features from neighboring intersections. Instead of isolated point predictions (like standard Random Forest regression), the GCN learns how the heat at node v_i is influenced by the physical infrastructure at an adjacent node v_j .

This structural mapping enables the system to generate a high-resolution Spatial SUHI Map, successfully bridging the gap between raw satellite radiometry and actionable, street-level urban planning intelligence.

CHAPTER III

Methodology

This chapter outlines the systematic workflow and technical procedures required to develop the spatial estimation framework. It details the comparative modeling approach, moving from data acquisition and spatial preprocessing through to graph construction, model training, and performance evaluation.

3.1 Research Design

This study employs a quantitative, comparative modeling design. It evaluates the efficacy of a non-Euclidean spatial machine learning model (Spatial Graph Convolutional Network) against a standard, non-structural baseline model (Random Forest regression) for estimating Surface Urban Heat Island (SUHI) intensity.

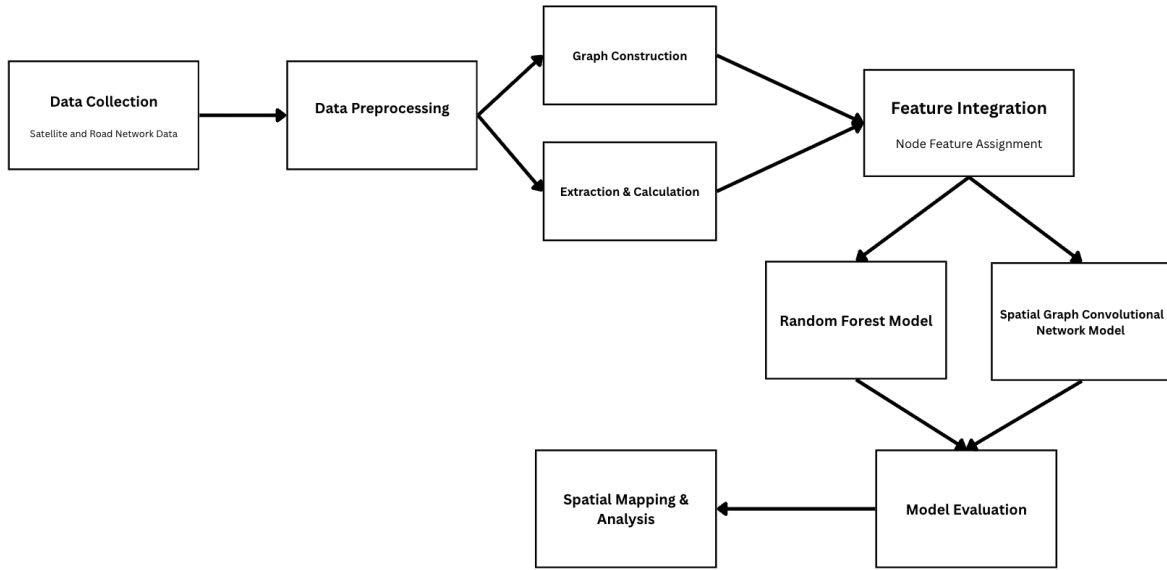


Figure 1. Methodology Framework

3.2 Data Collection

To construct a comprehensive spatial graph, this study utilizes a dual-source data acquisition strategy covering the dry seasons (March to May) of 2023 to 2025.

- **Satellite Remote Sensing Data:** Primary thermal and environmental variables will be derived from satellite imagery obtained from the USGS Earth Explorer archive.
 - **Landsat 8/9 OLI/TIRS:** Utilized for high-resolution multispectral data to extract surface infrastructure and vegetation indices.
 - **ECOSTRESS (Level-2):** Utilized to capture diurnal heat dynamics, allowing the model to account for both daytime peak heating and nighttime heat retention.
- **Vector Road Network Data:** Topological structural data for Davao City will be extracted from OpenStreetMap (OSM) using the Python OSMnx library. This vector data defines road classifications, intersection coordinates, and segment lengths to accurately map the physical connectivity of the city. The preliminary extraction of the unpruned, driveable urban core network yields a highly granular topology consisting of 26,546 intersection nodes ($|V|$) and 66,336 road segments ($|E|$).

3.3 Data Preprocessing

Raw data from these disparate sources will undergo rigorous preprocessing to ensure spatial alignment and structural integrity.

- **Image Calibration:** Satellite imagery will be radiometrically calibrated and atmospherically corrected to convert raw Digital Numbers (DN) into Top-of-Atmosphere (TOA) reflectance and brightness temperature.
- **Cloud Masking:** A cloud masking algorithm using the Quality Assessment (QA) Band will be applied to filter out atmospheric artifacts before cropping the images to the administrative boundaries of Davao City.
- **Topological Cleaning:** The OSM road network data will be cleaned to remove disconnected sub-graphs, isolate the main driving network, and simplify complex intersections, ensuring a mathematically viable graph structure for the deep learning model.

3.4 Feature Extraction and Graph Construction

This critical phase transforms the preprocessed data into a unified, mathematically structured dataset.

1. **Environmental Feature Computation:** Specific indices will be computed from the satellite imagery. To mitigate spatial autocorrelation and extreme buffer overlap within the highly dense residential street network, a tightly bounded 20-meter circular buffer is drawn around each intersection node. The median pixel value within this buffer is extracted for each variable. This statistical aggregation ensures the model captures the immediate microclimate of the street canyon while rejecting extreme surface albedo outliers (e.g., metal roof glare or shaded tree anomalies).
 - Target Variable: Land Surface Temperature (LST) to quantify SUHI.
 - Predictor Variables: Normalized Difference Built-up Index (NDBI) for impervious surface density, and Normalized Difference Vegetation Index (NDVI) for thermal buffering.
2. **Edge Weighting:** Connecting road segments (Edges) will be quantitatively weighted using segment length, road classification, and estimated traffic capacity. To handle

sparse and missing metadata native to crowdsourced OpenStreetMap data, a hierarchical lookup table is employed. This table maps OSM highway classifications to Philippine Department of Public Works and Highways (DPWH) and LTO baseline standards, imputing missing speed limits and lane counts to guarantee graph connectivity without data loss.

3. **Feature Integration:** The computed environmental features (LST, NDVI, NDBI) will be spatially mapped to the nearest road intersection (Nodes). This process outputs a fully Attributed Graph where the structural connectivity of the road network is explicitly linked with the thermal properties of the immediate environment.

3.5 Model Development

The study utilizes a comparative modeling approach to isolate and quantify the impact of incorporating road topology into thermal predictions.

- **Baseline Model (Random Forest Regression):** A Random Forest (RF) model will be trained on the isolated node features to serve as a control. This represents standard spatial machine learning methods that rely purely on tabular features (NDVI, NDBI) without structural awareness or topological connectivity.
- **Proposed Model (Spatial Graph Convolutional Network):** The Spatial GCN will be constructed to utilize the adjacency matrix of the road network. This architecture allows the model to aggregate features from neighboring nodes, effectively learning how heat at one intersection is structurally influenced by the conditions of physically connected road segments.

3.6 Evaluation and Spatial Mapping

The dataset will be partitioned into a standard **80/20 train-test split**. Both models will be evaluated on the unseen test nodes to measure their spatial interpolation capabilities.

Performance will be quantified using three standard regression error metrics:

- **Root Mean Square Error (RMSE):** To measure the standard deviation of the prediction errors, penalizing larger anomalies heavily.

- **Mean Absolute Error (MAE):** To measure the average magnitude of the errors without considering their direction.
- **Coefficient of Determination (R^2):** To assess the proportion of the variance in the LST that is predictable from the structural and environmental features.

Following evaluation, the superior model will be used to generate a high-resolution **Spatial SUHI Map** of Davao City, visualizing thermal distribution and pinpointing localized heat anomalies across the street network.

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